Approximating Orthogonal Functions with Orthogonal Polynomials

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Abstract

We can always approximate a function by a polynomial, and the approximation gets better as the degree of the polynomial goes up. The Hermite functions — the eigenfunctions of the Fourier transform — can be approximated by Hermite polynomials. The approximation coefficients form a triangular matrix with integer entries D_{mk}, which we conjecture to count weighted bicolour matchings. For the Chebyshev polynomials only two matchings survive, and the matrix is built from quartic factorials. We extend this construction to the Meixner–Pollaczek polynomials at $\lambda = \frac{1}{4}$ and $\frac{3}{4}$. In this case there are constants A and B such that every coefficient in the approximation matrix is of the form $sA + tB$ where s and t are integers. On the critical line, the Mellin transform carries the Hermite functions unitarily onto these two families.

1 The Construction

Each family follows the same pattern, in three steps.

Step 1: the polynomials. Fix a kernel $W \geq 0$ on a set X. It defines an inner product on polynomials,

$$\langle f, g\rangle_W = \int_X f\, g\, W.$$

Gram–Schmidt produces the orthogonal polynomials P_n:

$$\langle P_m, P_n\rangle_W = h_n \delta_{mn}.$$

Normalise:

$$\widehat{P}_n = P_n/\sqrt{h_n}.$$

Step 2: the functions. Multiplying by $\sqrt{W}$ turns them into the *associated functions*

$$\Phi_n = \widehat{P}_n\, \sqrt{W}.$$

These are orthonormal for the plain inner product:

$$\int_X \Phi_m\, \Phi_n = \delta_{mn}.$$

When $\sqrt{W}$ is not analytic, we instead split $W = w_+ w_-$ analytically and take the *dual pair of functions*

$$F_n = w_+\, \widehat{P}_n, \qquad G_n = w_-\, \widehat{P}_n, \qquad \int_X F_m\, G_n = \delta_{mn}.$$

Step 3: the coefficients. We seek *polynomial approximations of the functions.* The best degree-N approximation of Φ_m in $L^2(W)$ is given by

$$\sum_{k\le N} b_{mk}\,\widehat{P}_k.$$

The coefficients give a **triangular matrix**.

The coefficients are given by

$$b_{mk} = \langle \Phi_m, \widehat{P}_k\rangle_W = \int_X \left(\widehat{P}_m\sqrt{W}\right)\widehat{P}_k\,W = \int_X \widehat{P}_m\,\widehat{P}_k\,W^{3/2}.$$

In the dual case there are two coefficient matrices, one for each function of the pair:

$$c^+_{mk} = \langle F_m, \widehat{P}_k\rangle_W = \int_X \widehat{P}_m\,\widehat{P}_k\,w_+W \qquad (w_+W = w_+^2\,w_-),$$

$$c^-_{mk} = \langle G_m, \widehat{P}_k\rangle_W = \int_X \widehat{P}_m\,\widehat{P}_k\,w_-W \qquad (w_-W = w_-^2\,w_+).$$

2 Families of Orthogonal Polynomials

2.1 Hermite

In the Hermite polynomial case the kernel is given by $W = e^{-x^2}$ on $\mathbb{R}$, with $h_n = 2^n n!\sqrt{\pi}$. The Hermite functions are given by $\Phi_n = \widehat{H}_n e^{-x^2/2}$. We are interested in the coefficients of the polynomial approximation of the Hermite function by the Hermite polynomials

$$b_{mk} = \int \widehat{H}_m\widehat{H}_k\,e^{-3x^2/2}dx.$$

With $b_{00} = \sqrt{2/3}$, define the rescaled integers

$$D_{mk} \;=\; \frac{b_{mk}}{b_{00}}\,\sqrt{2^m m!\;2^k k!}\;\;3^{(m+k)/2} \;\in\; \mathbb{Z}.$$

$D_{mk} = 0$ for $m+k$ odd. The even entries ($m \le k \le 7$):

D_{mk}	$k=m$	$m{+}2$	$m{+}4$	$m{+}6$
$m=0$	1	−2	12	−120
1	4	−24	240	−3360
2	36	−408	6000	
3	528	−9120	181440	
4	10896	−254880		
5	290880	−8588160		
6	9529920			
7	369895680			

Theorem 2.1 (Hermite integrality)**.** *$D_{mk} \in \mathbb{Z}$, vanishing for $m+k$ odd. Explicitly, with $b = \frac{m-a}{2}$, $c = \frac{k-a}{2}$,*

$$D_{mk} \;=\; \sum_{a} 4^{a}\,(-2)^{b+c}\,a!\;\frac{m!}{a!\,b!\,2^{b}}\;\frac{k!}{a!\,c!\,2^{c}}\,,$$

and each factor $\dfrac{n!}{a!\,b!\,2^{b}}$ is an integer: it counts the partial matchings of $[n]$ with a unmatched points and b pairs.

Conjecture 2.2 (Hermite)**.** *D_{mk} is a weighted count of perfect matchings. Colour m points red and k points blue; then*

$$D_{mk} \;=\; \sum_{\substack{\text{perfect matchings}\\ \text{of the } m+k \text{ points}}} 4^{\#\{\text{red–blue pairs}\}}\,(-2)^{\#\{\text{same-colour pairs}\}}.$$

Theorem 2.3 (Hermite diagonal)**.** *The diagonal is manifestly positive:*

$$\frac{D_{nn}}{n!} \;=\; \sum_{j} \binom{n}{2j}\binom{2j}{j}\,4^{\,n-2j} \;=\; 1, 4, 18, 88, 454, 2424, \ldots \quad (\textit{OEIS A081671}),$$

the largest coefficient of $(1+4x+x^2)^n$, with ordinary generating function $1/\sqrt{(1-2x)(1-6x)}$ and exponential generating function $e^{4x}I_0(2x)$.

Reverse

The *reverse* expansion $\widehat{H}_m = \sum_k \widetilde{b}_{mk}\,\Phi_k$ has one power of $\sqrt{W}$ in place of three:

$$\widetilde{b}_{mk} = \int \widehat{H}_m \widehat{H}_k\, e^{-x^2/2}\,dx.$$

With $\widetilde{b}_{00} = \sqrt{2}$, the same rescaling without the power of 3,

$$S_{mk} \;=\; \frac{\widetilde{b}_{mk}}{\widetilde{b}_{00}}\,\sqrt{2^m m!\;2^k k!} \;\in\; \mathbb{N}.$$

$S_{mk} = 0$ for $m+k$ odd. The even entries ($m \le k \le 7$):

S_{mk}	$k=m$	$m{+}2$	$m{+}4$	$m{+}6$
$m=0$	1	2	12	120
1	4	24	240	3360
2	36	408	6000	
3	528	9120	181440	
4	10896	254880		
5	290880	8588160		
6	9529920			
7	369895680			

Theorem 2.4 (Hermite, reverse direction)**.** *$S_{mk} = (-1)^{(k-m)/2}\, D_{mk}$ — the forward matrix with the signs stripped. In the matchings of Conjecture 2.2 the same-colour weight -2 is replaced by $+2$, making the sum manifestly positive:*

$$S_{mk} \;=\; \sum_{\substack{\text{perfect matchings}\\ \text{of the } m+k \text{ points}}} 4^{\#\{\text{red–blue pairs}\}}\; 2^{\#\{\text{same-colour pairs}\}}.$$

2.2 Chebyshev

In the Chebyshev case the kernel is $W = (1-x^2)^{-1/2}$ on $(-1,1)$, with $h_0 = \pi$ and $h_n = \pi/2$. The associated functions are $\Phi_n = \widehat{T}_n\,(1-x^2)^{-1/4}$. Substituting $x = \cos\theta$ turns every matching integral into a two-term sum,

$$\int_{-1}^{1} T_m T_k\,(1-x^2)^{-3/4}\,dx \;=\; \tfrac{1}{2}\big[\,J(m-k) + J(m+k)\,\big], \qquad J(q) = \int_0^{\pi} \frac{\cos q\theta}{\sqrt{\sin\theta}}\,d\theta,$$

because $T_m T_k$ linearises into two cosines. Where Hermite sums over all matchings, Chebyshev keeps only two.

Let $P(n) = 1\cdot 5\cdots(4n-3)$ and $Q(n) = 3\cdot 7\cdots(4n-1)$ denote the quartic factorials. Then

$$J(0) = \frac{\sqrt{\pi}\,\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})}, \qquad J(2n) = J(0)\,\frac{P(n)}{Q(n)} \;>\; 0.$$

Every coefficient is a rational multiple of the single constant $J(0)$. With $s = \frac{m+k}{2}$, $d = \frac{|m-k|}{2}$, define

$$D_{mk} \;=\; \frac{Q(s)}{J(0)} \int_{-1}^{1} T_m\,T_k\,(1-x^2)^{-3/4}\,dx.$$

Theorem 2.5 (Chebyshev integrality)**.** *$D_{mk} = 0$ for $m+k$ odd, and otherwise*

$$D_{mk} \;=\; \frac{1}{2}\left[\,P(d)\,\frac{Q(s)}{Q(d)} + P(s)\,\right]$$

is a positive integer: both bracketed terms are products of odd numbers, so their sum is even.

The entries ($m \le k \le 7$):

D_{mk}	$k=m$	$m{+}2$	$m{+}4$	$m{+}6$
$m=0$	1	1	5	45
1	2	6	50	630
2	13	61	705	11385
3	138	870	12810	251910
4	2025	15945	284685	6592725
5	37890	356790	7477650	199153350
6	861525	9424485	226586025	6818954625
7	23052330	286937910	7779098250	260942613750

Theorem 2.6 (Chebyshev diagonal and edge)**.** *The edge is the quartic factorial, $D_{0,2n} = P(n)$ (OEIS A007696), and the diagonal is the average of the two quartic factorials,*

$$D_{nn} = \frac{Q(n) + P(n)}{2} \;=\; 1,\, 2,\, 13,\, 138,\, 2025,\, 37890,\, 861525,\, \ldots$$

This sequence is not in the OEIS.

Reverse

The reverse expansion uses $W^{1/2} = (1-x^2)^{-1/4}$ in place of $W^{3/2}$, hence $\widetilde{J}(q) = \int_0^\pi \cos q\theta \,\sqrt{\sin\theta}\; d\theta$. The third quartic class $\widetilde{P}(n) = 5 \cdot 9 \cdots (4n+1)$ appears, and the ratios turn negative,

$$\widetilde{J}(0) = \frac{\sqrt{\pi}\,\Gamma(\frac{3}{4})}{\Gamma(\frac{5}{4})}, \qquad \widetilde{J}(2n) = -\,\widetilde{J}(0)\,\frac{Q(n-1)}{\widetilde{P}(n)} \quad (n \geq 1).$$

Define $S_{mk} = \dfrac{\widetilde{P}(s)}{\widetilde{J}(0)} \displaystyle\int_{-1}^{1} T_m T_k\,(1-x^2)^{-1/4} dx$. Then $S_{mk} \in \mathbb{Z}$ by the same parity argument, with edge $S_{0,2n} = -Q(n-1)$ and diagonal $S_{nn} = \frac{1}{2}[\widetilde{P}(n) - Q(n-1)] = 1, 2, 21, 282, 4857, \ldots$ (also not in the OEIS). The entries ($m \leq k \leq 7$):

S_{mk}	$k = m$	$m{+}2$	$m{+}4$	$m{+}6$
$m = 0$	1	−1	−3	−21
1	2	−6	−30	−294
2	21	−69	−447	−5481
3	282	−1110	−8694	−126630
4	4857	−22617	−206955	−3474765
5	102690	−555030	−5804190	−110124630
6	2577645	−15898365	−186995655	−3951960705
7	74949210	−520103430	−6796188630	−158228220150

The sign pattern is inverted relative to Hermite: there the approximation matrix D carried the signs and the reverse matrix S was positive; here D is positive and S carries the signs. The three quartic residue classes share the work — P and Q build the forward matrix, Q and $\widetilde{P}$ the reverse.

The U-kernel

The kernel $W = (1-x^2)^{1/2}$ (Chebyshev-U) uses the fourth and last quartic class Q^+. Here $U_m U_k$ linearises into a *difference* of cosines, and with $K(q) = \int_0^\pi \cos q\theta \,\sin^{3/2}\theta\, d\theta$,

$$Q^+(n) = 7 \cdot 11 \cdots (4n+3), \qquad K(0) = \frac{J(0)}{3}, \qquad K(2n) = -\,3K(0)\,\frac{P(n-1)}{Q^+(n)} \quad (n \geq 1).$$

Define

$$D^U_{mk} \;=\; \frac{Q^+(s+1)}{K(0)} \int_0^\pi \sin\big((m{+}1)\theta\big)\sin\big((k{+}1)\theta\big)\sqrt{\sin\theta}\; d\theta.$$

Theorem 2.7 (U integrality). *$D^U_{mk} = 0$ for $m+k$ odd, and otherwise*

$$D^U_{mk} \;=\; \frac{1}{2}\left[\, Q^+(s{+}1)\,\rho(d) + 3P(s)\,\right] \;\in\; \mathbb{Z}, \qquad \rho(0)=1, \quad \rho(n) = -\,\frac{3P(n-1)}{Q^+(n)},$$

with edge $D^U_{0,2n} = -15\,P(n-1)$ for $n \geq 1$ (the quartic factorial A007696 again) and diagonal

$$D^U_{nn} = \frac{Q^+(n{+}1) + 3P(n)}{2} \;=\; 5,\ 40,\ 585,\ 11040,\ 253245,\ \ldots$$

not in the OEIS. The diagonal is positive and every other entry negative — a third sign pattern, after the all-positive D and the alternating S.

An asymmetric kernel: Catalan numbers

The kernel $W = (1-x)^{1/2}(1+x)^{-1/2}$ breaks the $x \mapsto -x$ symmetry; its orthogonal polynomials are the fourth-kind Chebyshev $W_n(\cos\theta) = \sin\big((n{+}\frac{1}{2})\theta\big)/\sin(\theta/2)$, with $h_n = \pi$ uniformly. The Wallis power drops to zero, the constant $\Gamma(\frac{1}{4})$ disappears, and the whole matching matrix lies in $\mathbb{Z}[\frac{1}{2}]\cdot\sqrt{2}$ — dyadic rationals. The first row is classical combinatorics: with $b_{0k} = \frac{1}{\pi}\int W_0 W_k W^{3/2}\,dx$,

$$2^k\sqrt{2}\;b_{0k} \;=\; \begin{cases} 2\binom{2n}{n} & k = 2n \geq 2, \\ -\,(4n+3)\,\mathrm{Cat}(n) & k = 2n+1, \end{cases} \qquad 2^0\sqrt{2}\;b_{00} = 3,$$

giving $3,\ -3,\ 4,\ -7,\ 12,\ -22,\ 40,\ -75,\ 140,\ -266, \ldots$ (verified exactly through $k = 13$; the signed sequence is not in the OEIS). The quartic weights of the symmetric kernels cannot count lattice paths; the asymmetric kernel apparently can.

In total the Chebyshev-type matrices use the four quartic factorial classes $1{\cdot}5{\cdot}\cdots$, $3{\cdot}7{\cdot}\cdots$, $5{\cdot}9{\cdot}\cdots$, $7{\cdot}11{\cdot}\cdots$ — the four odd arithmetic progressions of difference four, each exactly once.

2.3 Laguerre

For the Laguerre polynomials with $\alpha = 0$ the kernel is $W = e^{-x}$ on $(0,\infty)$. We denote the Laguerre polynomials by L_n and the Laguerre functions by $\Phi_n = L_n e^{-x/2}$. They are already orthonormal ($h_n = 1$). The coefficients

$$b_{mk} = \int_0^\infty L_m L_k\, e^{-3x/2} dx,$$

are all exactly rational. We rescale:

$$b_{mk} \;=\; \frac{2\,E_{mk}}{3^{\,m+k+1}}, \qquad E_{mk} \in \mathbb{N}:$$

E_{mk}	$k=0$	1	2	3	4	5
$m=0$	1	1	1	1	1	1
1		5	9	13	17	21
2			33	73	129	201
3				245	593	1181
4					1921	4881
5						15525

Theorem 2.8 (Laguerre)**.** *E_{mk} counts weighted Delannoy paths:*

$$E_{mk} \;=\; \sum_{\text{paths } (0,0)\to(m,k)} 3^{\#\text{diagonal steps}} \;=\; \sum_{j\ge 0} \binom{m}{j}\binom{m+k-j}{m}\, 3^{j},$$

the sum over monotone lattice paths with steps $(1,0)$, $(0,1)$, $(1,1)$, a diagonal step weighing 3. In particular $E_{1,k} = 4k+1$ and $E_{nn} = \sum_j \binom{n}{j}^2 4^j = {}_2F_1(-n,-n;1;4)$.

Reverse

The *reverse* expansion $L_m = \sum_k \widetilde{b}_{mk}\,\Phi_k$ gives the same Delannoy numbers:

$\widetilde{b}_{mk}$	$k=0$	1	2	3	4	5
$m=0$	2	−2	2	−2	2	−2
1		10	−18	26	−34	42
2			66	−146	258	−402
3				490	−1186	2362
4					3842	−9762
5						31050

Theorem 2.9 (Laguerre, reverse direction)**.**

$$\widetilde{b}_{mk} = \int_0^\infty L_m L_k\, e^{-x/2}\, dx \;=\; 2\,(-1)^{m+k}\, E_{mk}.$$

Reversing the direction strips the power of 3 and leaves a sign.

3 Not quite combinatorial

3.1 Meixner–Pollaczek ($\lambda = \frac{1}{4}$): a dual pair

For the Meixner–Pollaczek polynomials with $\lambda = \frac{1}{4}$ the kernel is $W = \Gamma(\frac{1}{4}+ix)\,\Gamma(\frac{1}{4}-ix)$ on $\mathbb{R}$, with $h_n = h_0\binom{2n}{n}/4^n$, $h_0 = \sqrt{2}\,\pi^{3/2}$. We denote the Meixner–Pollaczek polynomials by P_n. Here $\sqrt{W}$ has branch points at $\pm\frac{i}{4}$; the analytic split is $w_\pm = \Gamma(\frac{1}{4}\pm ix)$ and the dual pair of functions is given by

$$F_n = \Gamma(\tfrac{1}{4}+ix)\,\widehat{P}_n, \qquad G_n = \Gamma(\tfrac{1}{4}-ix)\,\widehat{P}_n, \qquad \int_{\mathbb{R}} F_m G_n\, dx = \delta_{mn}.$$

The coefficients of the approximation matrix are given by:

$$c^+_{mk} = \int \widehat{P}_m \widehat{P}_k\, \Gamma(\tfrac{1}{4}+ix)^2\, \Gamma(\tfrac{1}{4}-ix)\, dx, \qquad c^-_{mk} = \int \widehat{P}_m \widehat{P}_k\, \Gamma(\tfrac{1}{4}+ix)\, \Gamma(\tfrac{1}{4}-ix)^2\, dx$$

They are complex conjugates of each other, $c^-_{mk} = \overline{c^+_{mk}}$. With the rescaling

$$T_{mk} = \frac{2^m m!\; 2^k k!}{h_0} \int_{\mathbb{R}} P_m P_k\, \Gamma(\tfrac{1}{4}+ix)^2 \Gamma(\tfrac{1}{4}-ix)\, dx, \qquad c^+_{mk} = \frac{T_{mk}}{m!\,k!\,\sqrt{\binom{2m}{m}\binom{2k}{k}}}.$$

Up to a factor of i when $m+k$ is odd, every entry lies in $\mathbb{Z}A + \mathbb{Z}B$, where

$$A = \sqrt{2e}\, K_0(\tfrac{1}{2}), \qquad B = \sqrt{2e}\, K_1(\tfrac{1}{2}),$$

and the K_ν are the modified Bessel functions of the third kind

$$K_\nu(z) = \int_0^\infty e^{-z\cosh t}\cosh(\nu t)\,dt\,:$$

(m,k)	T_{mk}	(m,k)	T_{mk}
$(0,0)$	A	$(1,3)$	$2575A - 1440B$
$(0,1)$	$3A - 2B$	$(1,4)$	$39411A - 21978B$
$(0,2)$	$12B - 23A$	$(1,5)$	$394932B - 707389A$
$(0,3)$	$138B - 241A$	$(2,2)$	$2369A - 1320B$
$(0,4)$	$3265A - 1800B$	$(2,3)$	$33743A - 18834B$
$(0,5)$	$52907A - 29706B$	$(2,4)$	$317892B - 569719A$
$(1,1)$	$12B - 21A$	$(2,5)$	$6190602B - 11092229A$
$(1,2)$	$114B - 205A$	$(3,3)$	$295572B - 529549A$
		$(3,4)$	$5399634B - 9675553A$

Duality

The approximation of the other function of the pair, computed directly,

$$T^{-}_{mk} = \frac{2^m m!\,2^k k!}{h_0}\int_{\mathbb{R}} P_m P_k\,\Gamma(\tfrac14 + ix)\,\Gamma(\tfrac14 - ix)^2\,dx \;=\; \overline{T_{mk}}\,,$$

lies in the same lattice $\mathbb{Z}A + \mathbb{Z}B$. Conjugation flips the suppressed i to $-i$; at the fixed phase of the T-table the odd-parity entries change sign:

(m,k)	T^{-}_{mk}	(m,k)	T^{-}_{mk}
$(0,0)$	A	$(1,3)$	$2575A - 1440B$
$(0,1)$	$2B - 3A$	$(1,4)$	$21978B - 39411A$
$(0,2)$	$12B - 23A$	$(1,5)$	$394932B - 707389A$
$(0,3)$	$241A - 138B$	$(2,2)$	$2369A - 1320B$
$(0,4)$	$3265A - 1800B$	$(2,3)$	$18834B - 33743A$
$(0,5)$	$29706B - 52907A$	$(2,4)$	$317892B - 569719A$
$(1,1)$	$12B - 21A$	$(2,5)$	$11092229A - 6190602B$
$(1,2)$	$205A - 114B$	$(3,3)$	$295572B - 529549A$
		$(3,4)$	$9675553A - 5399634B$

Reverse

The *reverse* expansion $\widehat{P}_m = \sum_k D_{mk}F_k$ has one-Gamma coefficients

$$D_{mk} = \int \widehat{P}_m \widehat{P}_k\,\Gamma(\tfrac14 - ix)\,dx$$

over the single generator e^{-1} (a factor of i suppressed when $m+k$ is odd):

$$S_{mk} = 2^m m!\,2^k k!\,\frac{e}{2\pi}\int_{\mathbb{R}} P_m P_k\,\Gamma(\tfrac14 - ix)\,dx \;\in\; \mathbb{Z}\,:$$

S_{mk}	$k=0$	1	2	3	4	5
$m=0$	1	3	5	11	249	−2621
1		7	47	399	−2005	263
2			473	−617	22109	−402065
3				31511	−412557	−3017969
4					−1916623	−29235989
5						−1012892921

Definition 3.1 (Defect–chain matching)**.** A *defect–chain matching* is a combinatorial object richer than a matching. Take two rows of labelled points, $[m]$ and $[k]$. Within each row, some points are set aside as *defects*; the remaining points are arranged into a forest of nonempty ordered *chains*. Finally, chains in the $[m]$-row are partially matched with chains in the $[k]$-row. Each part carries a weight, and S_{mk} is the signed sum over all such configurations.

Theorem 3.2 (Meixner–Pollaczek $\frac{1}{4}$, reverse direction)**.** $S_{mk} = (-1)^{\lfloor (m+k)/2 \rfloor} \sum \mathrm{wt}$, *summed over the defect–chain matchings of* $[m]$, $[k]$*; the weights are* $(-1)^d(2d-1)!!$ *per defect set of size* d, $(-1)^{\ell-1}2^{\ell+1}$ *per chain of length* ℓ*, and* -1 *per matched pair of chains. Equivalently, with Lah numbers* $L(n,c) = \binom{n-1}{c-1} n!/c!$,

$$S_{mk} = (-1)^{\lfloor \frac{m+k}{2} \rfloor} \sum_{d_1,c_1,d_2,c_2} W_m(d_1,c_1)\, W_k(d_2,c_2) \sum_j (-1)^j \tbinom{c_1}{j} \tbinom{c_2}{j} j!\,,$$

$W_n(d,c) = \binom{n}{d}(-1)^d(2d-1)!!\; L(n-d,c)\,(-1)^{n-d-c}\,2^{n-d+c}$.

Reverse dual

The expansion over the other dual function, $\widehat{P}_m = \sum_k \widetilde{D}_{mk}\, G_k$ with $\widetilde{D}_{mk} = \int \widehat{P}_m \widehat{P}_k\, \Gamma(\frac{1}{4}+ix)\, dx$, obeys two symmetries: conjugation ($\widehat{P}$ is real on $\mathbb{R}$) gives $\widetilde{D}_{mk} = \overline{D_{mk}}$, and the substitution $x \mapsto -x$ gives $\widetilde{D}_{mk} = (-1)^{m+k} D_{mk}$. Together they force D_{mk} to be real for $m+k$ even and purely imaginary for $m+k$ odd — this is the origin of the suppressed factor of i. Computing directly, at the fixed phase i,

$$\widetilde{S}_{mk} = 2^m m!\; 2^k k!\; \frac{e}{2\pi} \int_{\mathbb{R}} P_m P_k\, \Gamma(\tfrac{1}{4}+ix)\, dx \;\in\; \mathbb{Z}\,:$$

$\widetilde{S}_{mk}$	$k=0$	1	2	3	4	5
$m=0$	1	−3	5	−11	249	2621
1		7	−47	399	2005	263
2			473	617	22109	402065
3				31511	412557	−3017969
4					−1916623	29235989
5						−1012892921

Theorem 3.3 (Meixner–Pollaczek $\frac{1}{4}$, reverse dual)**.** $\widetilde{S}_{mk} = (-1)^{m+k} S_{mk} = (-1)^{\lceil (m+k)/2 \rceil} \sum \mathrm{wt}$, *summed over the same defect–chain matchings, with the same weights, as Theorem 3.2: only the floor becomes a ceiling.*

3.2 Meixner–Pollaczek ($\lambda = \frac{3}{4}$): a dual pair

For the Meixner–Pollaczek polynomials with $\lambda = \frac{3}{4}$ the kernel is $W = \Gamma(\frac{3}{4} + ix)\,\Gamma(\frac{3}{4} - ix)$ on $\mathbb{R}$, with $h_n = h_0\,\Gamma(n + \frac{3}{2})/(n!\,\Gamma(\frac{3}{2}))$, $h_0 = (\pi/2)^{3/2}$. We denote the Meixner–Pollaczek polynomials by Q_n. Here $\sqrt{W}$ has branch points at $\pm\frac{3i}{4}$; the analytic split is $w_\pm = \Gamma(\frac{3}{4} \pm ix)$ and the dual pair of functions is given by

$$F_n = \Gamma(\tfrac{3}{4} + ix)\,\widehat{Q}_n, \qquad G_n = \Gamma(\tfrac{3}{4} - ix)\,\widehat{Q}_n, \qquad \int_{\mathbb{R}} F_m G_n\,dx = \delta_{mn}.$$

The coefficients of the approximation matrix are given by:

$$c^+_{mk} = \int \widehat{Q}_m \widehat{Q}_k\,\Gamma(\tfrac{3}{4} + ix)^2\,\Gamma(\tfrac{3}{4} - ix)\,dx, \qquad c^-_{mk} = \int \widehat{Q}_m \widehat{Q}_k\,\Gamma(\tfrac{3}{4} + ix)\,\Gamma(\tfrac{3}{4} - ix)^2\,dx$$

They are complex conjugates of each other, $c^-_{mk} = \overline{c^+_{mk}}$. With the rescaling

$$\mathcal{U}_{mk} = \frac{2^{m+k} m!\,k!}{\pi^2/2}\,\mathrm{Re}\big[(-i)^{m+k} J_{mk}\big], \qquad J_{mk} = \int Q_m Q_k\,\Gamma(\tfrac{3}{4} + ix)^2 \Gamma(\tfrac{3}{4} - ix)\,dx,$$

every entry lies in $\mathbb{Z}A' + \mathbb{Z}B'$, where

$$A' = U(\tfrac{3}{2}, 1, 1) \approx 0.35793, \qquad B' = U(\tfrac{5}{2}, 2, 1) \approx 0.21533,$$

U being Tricomi's confluent hypergeometric function (no Bessel simplification: $\lambda = \frac{1}{4}$ is the special Kummer–Bessel locus):

(m,k)	$\mathcal{U}_{mk}$	(m,k)	$\mathcal{U}_{mk}$
$(0,0)$	A'	$(1,3)$	$7035A' - 11760B'$
$(0,1)$	$3A' - 6B'$	$(1,4)$	$128643A' - 213630B'$
$(0,2)$	$39A' - 60B'$	$(2,2)$	$6609A' - 10920B'$
$(0,3)$	$513A' - 870B'$	$(2,3)$	$112695A' - 187350B'$
$(0,4)$	$8673A' - 14280B'$	$(2,4)$	$2215047A' - 3679860B'$
$(1,1)$	$33A' - 60B'$	$(3,3)$	$2078169A' - 3456420B'$
$(1,2)$	$453A' - 750B'$	$(3,4)$	$43522881A' - 72346830B'$

Duality

The approximation of the other function of the pair, computed directly at the fixed phase of the $\mathcal{U}$-table,

$$\mathcal{U}^-_{mk} = \frac{2^{m+k} m!\,k!}{\pi^2/2}\,\mathrm{Re}\big[(-i)^{m+k}\,\overline{J_{mk}}\big],$$

again negates exactly the odd-parity entries; with the phase conjugated to i^{m+k} the $\mathcal{U}$-table reappears identically:

(m,k)	$\mathcal{U}^-_{mk}$	(m,k)	$\mathcal{U}^-_{mk}$
$(0,0)$	A'	$(1,3)$	$7035A' - 11760B'$
$(0,1)$	$6B' - 3A'$	$(1,4)$	$213630B' - 128643A'$
$(0,2)$	$39A' - 60B'$	$(2,2)$	$6609A' - 10920B'$
$(0,3)$	$870B' - 513A'$	$(2,3)$	$187350B' - 112695A'$
$(0,4)$	$8673A' - 14280B'$	$(2,4)$	$2215047A' - 3679860B'$
$(1,1)$	$33A' - 60B'$	$(3,3)$	$2078169A' - 3456420B'$
$(1,2)$	$750B' - 453A'$	$(3,4)$	$72346830B' - 43522881A'$

Reverse

The *reverse* expansion $\widehat{Q}_m = \sum_k D_{mk} F_k$ has one-Gamma coefficients

$$D_{mk} = \int \widehat{Q}_m \widehat{Q}_k \, \Gamma(\tfrac{3}{4} - ix)\, dx$$

over the single generator e^{-1} (a phase $(-i)^{|m-k|}$ suppressed):

$$S_{mk} = 2^m m!\; 2^k k!\; \frac{e}{2\pi} \int Q_m Q_k \, \Gamma(\tfrac{3}{4} - ix)\, dx \;\in\; \mathbb{Z}:$$

S_{mk}	$k=0$	1	2	3	4	5
$m=0$	1	-1	-9	-43	-95	1671
1		15	23	-283	-4767	-52649
2			529	5475	31703	-65119
3				-15145	363979	8950653
4					-8709567	-100346585
5						123082079

Theorem 3.4 (Meixner–Pollaczek $\frac{3}{4}$, reverse direction)**.** *$S_{mk} \in \mathbb{Z}$ for all m, k, a signed sum over the same defect–chain matchings as Theorem 3.2: for $m \le k$,*

$$S_{mk} = (-1)^k \sum_{d_1,c_1,d_2,c_2} W'_m(d_1,c_1)\, W'_k(d_2,c_2) \sum_j (-1)^j \tbinom{c_1}{j} \tbinom{c_2}{j} j!\,,$$

$$W'_n(d,c) = \binom{n}{d} (-1)^d (2d+1)!!\; L(n-d,c)\, (-1)^{n-d-c}\, 2^{n-d+c}.$$

Only the defect weight changes, from $(2d-1)!!$ to $(2d+1)!!$: the Charlier kernel $e^{-\sigma-\tau+\sigma\tau}$ is universal in λ, and the Jacobian exponent shifts the odd double factorial by the half-integer λ.

Reverse dual

As for $\lambda = \frac{1}{4}$, the expansion over the other dual function is the complex conjugate; computed directly, at the fixed phase $(-i)^{|m-k|}$,

$$\widetilde{S}_{mk} = 2^m m!\; 2^k k!\; \frac{e}{2\pi} \int Q_m Q_k \, \Gamma(\tfrac{3}{4} + ix)\, dx \;\in\; \mathbb{Z}:$$

$\widetilde{S}_{mk}$	$k=0$	1	2	3	4	5
$m=0$	1	1	-9	43	-95	-1671
1		15	-23	-283	4767	-52649
2			529	-5475	31703	65119
3				-15145	-363979	8950653
4					-8709567	100346585
5						123082079

Theorem 3.5 (Meixner–Pollaczek $\frac{3}{4}$, reverse dual)**.** $\widetilde{S}_{mk} = (-1)^{m+k}\, S_{mk}$*: the signed sum of Theorem 3.4 with the prefactor $(-1)^k$ replaced by $(-1)^m$ (for $m \le k$), the weights W' unchanged.*

4 Automata

Each generating function below is a composition: an algebraic prefactor times $g(\zeta(w,u))$, where g is the kernel of the family and ζ an inner function. Extracting the coefficient of $w^m u^k$ applies ∂_w m times and ∂_u k times, then evaluates at $w = u = 0$.
Watch the derivatives land on $g(\zeta)$ one at a time:

$$\partial_1\, g(\zeta) = g'(\zeta)\,\partial_1\zeta, \qquad \partial_2\partial_1\, g(\zeta) = g''(\zeta)\,\partial_2\zeta\,\partial_1\zeta + g'(\zeta)\,\partial_2\partial_1\zeta.$$

At each step the new derivative either hits g again, opening a new inner factor — a new *block* — or deepens an existing one. Iterating, the terms are indexed by the set partitions π of the $m+k$ derivative slots (Faà di Bruno):

$$\partial^{m+k} g(\zeta) = \sum_{\pi} g^{(|\pi|)}(\zeta) \prod_{B\in\pi} \partial_B\,\zeta.$$

Each term factors in two ways. For each block B the factor $\partial_B\,\zeta$ is a mixed partial of the *inner* function at the base point — a rational constant, essentially a Taylor coefficient of ζ. So the shape of the partition — which slots are grouped with which — only multiplies the term by a rational dressing. The outer function is differentiated once per block, giving $g^{(j)}$ with $j = |\pi|$, and $g^{(j)}$ is what the automaton computes. Of all the information in π the automaton therefore sees only the single number j: two partitions with the same number of blocks feed it identically, differing only in their rational dressing.
The walks compute $g^{(j)}$.
When the kernel satisfies a linear ODE of order r, its derivatives close on r states, and each application of d/dz is an $r \times r$ transfer matrix; expanding the j-fold matrix product over intermediate states is a sum over walks of length j in the r-state automaton, one step per block, each step weighted by an entry of the transfer matrix.

Two refinements: derivatives may also land on the algebraic prefactor, distributed by the Leibniz rule before the partition is formed; and the matrix entries are z-dependent, so repeated differentiation also dresses the walk steps with their derivatives. Both contribute rational factors.

A toy example

Take the Meixner–Pollaczek entry J_{11}: two derivative slots on

$$G = \frac{\sqrt{2}\,h_0\; g(\zeta)}{\sqrt{(1-iw)(1-iu)}}, \qquad g(z) = e^{z/2}K_0(z/2), \quad \zeta = \varphi(w)\,\varphi(u).$$

At the base point $w = u = 0$: $\zeta = 1$ and

$$\partial_w\zeta = \partial_u\zeta = 2i, \qquad \partial_w\partial_u\zeta = -4,$$

while a slot landing on the prefactor contributes $i/2$. The walk states carry the values $\sqrt{2}\,e^{\zeta/2}K_0(\zeta/2) = A$ and $\sqrt{2}\,e^{\zeta/2}K_1(\zeta/2) = B$ at $\zeta = 1$. Each $d/d\zeta$ steps $K_0 \to K_0$ with weight $\frac{1}{2}$ (the exponential differentiating) or $K_0 \to K_1$ with weight $-\frac{1}{2}$; from K_1, to K_0 with weight $-\frac{1}{2}$ or the self-loop with weight $\frac{1}{2} - 1/\zeta = -\frac{1}{2}$ — the automaton of the next subsection, dressed by the exponential and the chain-rule halves. In units of h_0:

slots	blocks	walk	contributes
both on prefactor	—	empty walk	$-\frac{1}{4}A$
one on g ($\times 2$)	$\{w\}$ or $\{u\}$	$K_0 \to K_0$	$-A$
		$K_0 \to K_1$	$+B$
both on g, split	$\{w\}\{u\}$	$K_0 \to K_0 \to K_0$, $K_0 \to K_1 \to K_0$	$-2A$
		$K_0 \to K_0 \to K_1$, $K_0 \to K_1 \to K_1$	0
both on g, merged	$\{wu\}$	$K_0 \to K_0$	$-2A$
		$K_0 \to K_1$	$+2B$

The split partition walks two steps with Taylor factor $\partial_w\zeta\,\partial_u\zeta = -4$; the merged block walks one step and picks up $\partial_w\partial_u\zeta = -4$ instead — the Cayley transform makes the two coincide, but they weight walks of different lengths. Total: $3B - \frac{21}{4}A$, i.e. $J_{11} = h_0\,(12B - 21A)/4$ — the entry $T_{11} = 12B - 21A$ of the table in §3.1.

4.1 Meixner–Pollaczek ($\lambda = \frac{1}{4}$): a dual pair

The unrescaled coefficients $J_{mk} = h_0\,T_{mk}/(2^m m!\,2^k k!)$ have a closed generating function: with

$$\zeta(w,u) = \varphi(w)\,\varphi(u), \qquad \varphi(t) = \frac{1+it}{1-it} \quad \text{(the Cayley transform)},$$

$$\sum_{m,k\geq 0} J_{mk}\,w^m u^k = \frac{2\pi^{3/2}\,e^{\zeta/2}\,K_0(\zeta/2)}{\sqrt{(1-iw)(1-iu)}}.$$

Conjecture 4.1 (Meixner–Pollaczek $\frac{1}{4}$, forward direction)**.** *$T_{mk} \in \mathbb{Z}A \oplus \mathbb{Z}B$ for all m, k. No explicit combinatorial formula is known.*

The automaton has states $\{K_0, K_1\}$ for the kernel $g(z) = e^{z/2}K_0(z/2)$, with transitions given by the Bessel relations

$$K_0' = -K_1, \qquad K_1' = -K_0 - \frac{K_1}{z},$$

evaluated at $z = \frac{1}{2}$: a step either switches state (weight -1) or self-loops at K_1 (weight $-1/z = -2$). Walks start at K_0; those ending at K_0 contribute to the coefficient of A, those ending at K_1 to that of B.

The same automaton covers the other function of the pair: conjugating the generating function is the substitution $(w,u) \mapsto (-w,-u)$ — the Cayley transform obeys $\varphi(-t) = \overline{\varphi(t)}$ — which leaves the kernel, states and weights unchanged and only flips the sign of the odd-parity dressings, reproducing the duality of §3.1.

4.2 Meixner–Pollaczek ($\lambda = \frac{3}{4}$): a dual pair

The coefficients J_{mk} have a closed generating function: with ζ as above,

$$\sum_{m,k\geq 0} J_{mk}\,w^m u^k = \frac{\pi^2}{2}\,\frac{U\left(\frac{3}{2},\,1,\,\zeta\right)}{(1-iw)^{3/2}(1-iu)^{3/2}}.$$

This is the $\lambda = \frac{1}{4}$ generating function with the exponent $\frac{1}{2}$ promoted to $\frac{3}{2}$ and the Bessel kernel $e^{\zeta/2}K_0(\zeta/2)$ replaced by the Tricomi kernel $U(\frac{3}{2},1,\zeta)$.

Conjecture 4.2 (Meixner–Pollaczek $\frac{3}{4}$, forward direction)**.** *$\mathcal{U}_{mk} \in \mathbb{Z}A' \oplus \mathbb{Z}B'$ for all m, k. No explicit combinatorial formula is known.*

The automaton has states $\{U_0, U_1\}$ for the kernel $U_0 = U(\frac{3}{2}, 1, z)$, $U_1 = U(\frac{5}{2}, 2, z)$, with transitions given by the Tricomi contiguous relations

$$U_0' = -\tfrac{3}{2}\, U_1, \qquad U_1' = -\frac{U_0}{z} + \frac{z-1}{z}\, U_1,$$

evaluated at $z = 1$: the step $U_0 \to U_1$ has weight $-\frac{3}{2}$, the step $U_1 \to U_0$ has weight -1, and the self-loop on U_1 has weight 0. Walks start at U_0; those ending at U_0 contribute to the coefficient of $A' = U_0(1)$, those ending at U_1 to that of $B' = U_1(1)$. As for $\lambda = \frac{1}{4}$, the other function of the pair is covered by the same automaton via $(w, u) \mapsto (-w, -u)$.

5 The Mellin Transform and the Riemann Zeta Function

The Mellin transform sends the Hermite functions to the Meixner–Pollaczek dual functions. The Mellin transform is the Fourier transform of the multiplicative group $\mathbb{R}_+$; on the critical line $s = \frac{1}{2} + 2ix$ it is unitary.

Theorem 5.1. *For every $n \geq 0$, with $m = \lfloor n/2 \rfloor$,*

$$\mathcal{M}\big[\Phi_n(\sqrt{2}\,\cdot)\big]\big(\tfrac{1}{2} + 2ix\big) = \frac{2^{1/4}\sqrt{\pi}}{2}\; i^m\, F_m(x),$$

where F_m belongs to the $\lambda = \frac{1}{4}$ family for n even and to the $\lambda = \frac{3}{4}$ family for n odd.

One constant serves every n and both parities. Its modulus is forced: $|c|^2 = \pi/2\sqrt{2}$ is the Plancherel constant of the $\sqrt{2}$-dilated Mellin transform, so the map is unitary, and i^m is its eigenvalue phase — the Mellin analogue of the Fourier eigenvalues $(-i)^n$ of Φ_n. Underneath is a polynomial identity: the Bump–Ng polynomials (Bump–Ng 1986) at $s = \frac{1}{2} + 2ix$ *are* the Meixner–Pollaczek polynomials (Coffey 2013), and their real-zero property ("local RH") is Meixner–Pollaczek orthogonality. The kernel $\Gamma(\frac{1}{4} + ix)\,\Gamma(\frac{1}{4} - ix)$ is $|\Gamma(s/2)|^2$, the archimedean factor of the completed zeta function $\xi(s)$; the functional equation $s \mapsto 1 - s$ is $x \mapsto -x$, which swaps $F \leftrightarrow G$ — the dual pair *is* the functional equation, seen locally.

One family on $\mathbb{R}$, two on $\mathbb{R}_+$: Mellin lives on the half-line, where even and odd are no longer related by $x \mapsto -x$, and the Hermite family splits into two independent halves — $\Gamma(s/2)$ for even, $\Gamma(\frac{s+1}{2})$ for odd, the archimedean factors of even and odd Dirichlet L-functions. The even Hermite functions alone are a complete orthogonal system on $\mathbb{R}_+$ (an even function is determined by its half), and their Mellin image is the complete system $\{F_n\}$; the biorthogonality $\int_{\mathbb{R}} F_m G_n\, dx = \delta_{mn}$ of §3 is Hermite orthogonality on $\mathbb{R}_+$ by Plancherel. What is lost is the orthogonality between the halves: on the half-line Φ_{2m} and Φ_{2n+1} are no longer orthogonal, and their cross-pairing is the unbalanced kernel $\Gamma(\frac{1}{4} + ix)\,\Gamma(\frac{3}{4} - ix)$, which we treat elsewhere.

So the two Meixner–Pollaczek families are not an extension of Hermite: they *are* the Hermite family, viewed multiplicatively. The matrices of this paper change basis between a function and its own Mellin shadow — and why that produces integers, and lattices $\mathbb{Z}A \oplus \mathbb{Z}B$, we do not understand.

6 Proofs Available Upon Request